

Chapter 1

Dataset Understanding

1 .Introduction

This report presents a comprehensive analysis of the Dirty Cafe Sales Dataset obtained from Kaggle. The dataset simulates realworld messy transactional data from a cafe, and includes various data quality issues such as missing values, inconsistent entries, and outliers.

The objective of this homework is to understand the dataset structure, perform exploration data analysis, apply statistical descriptions, and carry out proper data cleaning to prepare the data for future use in data mining tasks

1.1 Dataset

The raw dataset contains 10,000 transaction records with 8 attributes. After data cleaning, 8,788 records were retained for analysis. Below is a description of each attribute:

Attribute	Data Type	Description
Transaction ID	Nominal	Unique identifier for each transaction
Item	Nominal	Name of the purchased item (e.g., Coffee, Salad, Cake)
Quantity	Numerical – Discrete	Number of units purchased (range: 1–5)
Price Per Unit	Numerical – Continuous	Price of a single unit (range: 1.0–5.0)
Total Spent	Numerical – Continuous	Total transaction amount (Quantity × Price Per Unit)
Payment Method	Nominal	Payment method used (e.g., Cash, Credit Card)

Location	Nominal	Location where the transaction was recorded
Attribute	Data Type	Description
Transaction Date	Continuous – Date/Time	Date and time of the transaction (year 2023)

Chapter 2

Data Exploratory

2.1 Business Application

Sales Forecasting

By analyzing Total Spent trends over time (Transaction Date), the cafe can predict next month's revenue and adjust inventory and staffing accordingly. Key findings support this application:

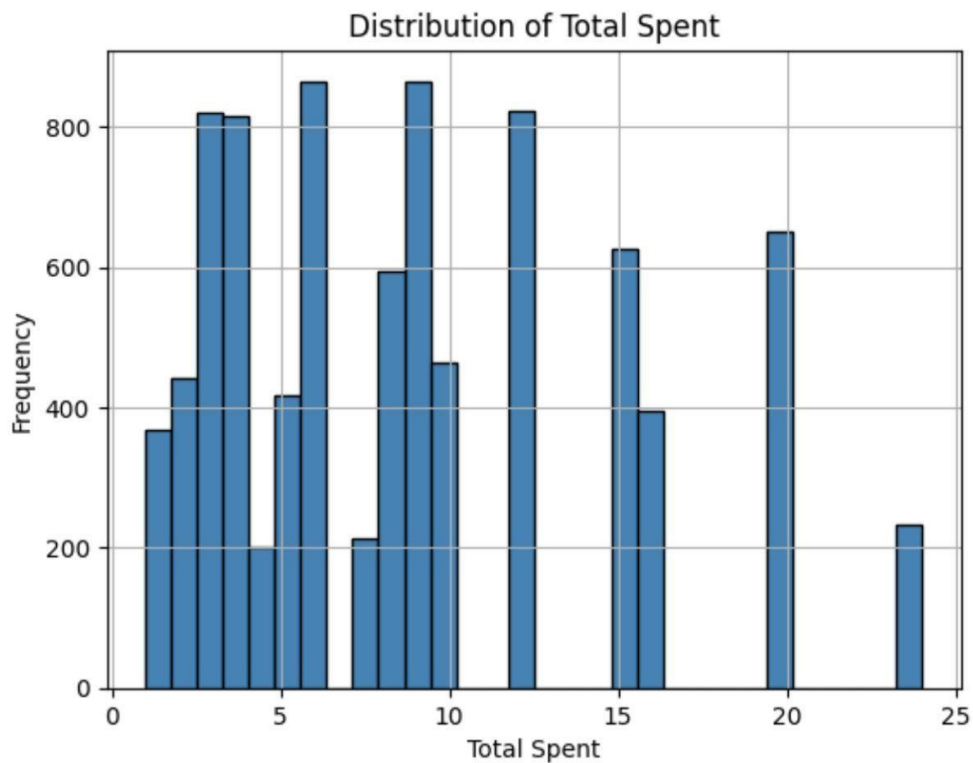
- October recorded the highest monthly sales (~6,800), while February had the lowest (~6,100).
- Weekends (Friday, Saturday, Sunday) consistently show peak sales activity.
- Salad dominates as the top-selling item throughout the year.

These patterns allow the business to plan targeted promotions during low-performing periods and optimize resources during highdemand periods.

2.2 Exploration Visualizations

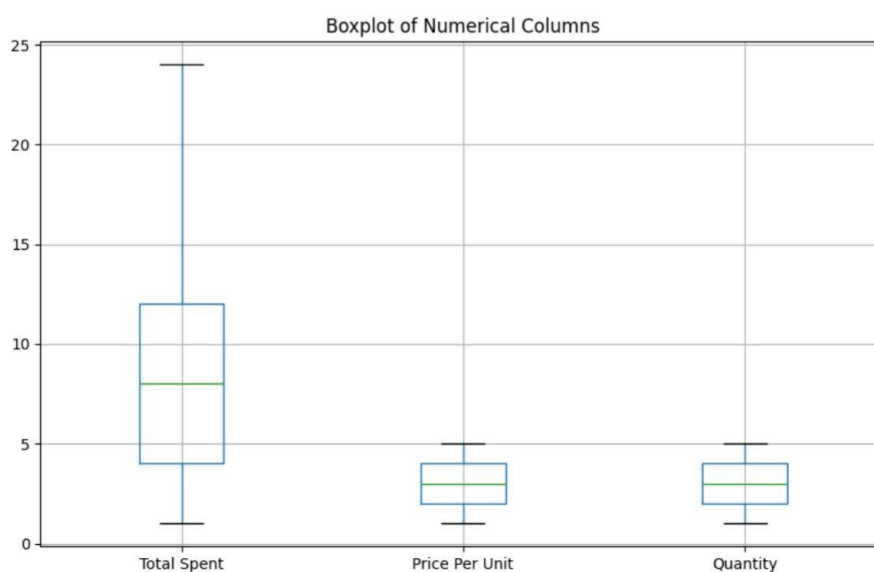
Visualization 1 – Distribution of Total Spent (Histogram)

- Most customers spend between 4 and 12 units per transaction.
- The distribution is roughly uniform with no dominant single peak.
- Very few customers spend above 20 units.
- This suggests cafe prices are accessible and affordable for a wide customer base.



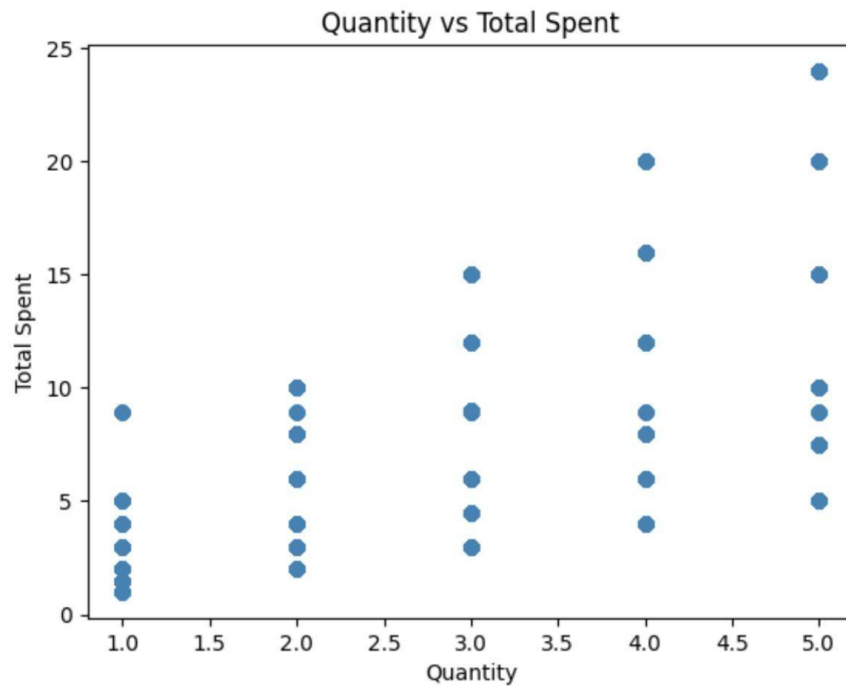
Visualization 2 – Boxplot of Numerical Columns

- Total Spent has a median of approximately 8, ranging from 1 to 24.
- Price Per Unit and Quantity are evenly distributed between 1 and 5.
- After data cleaning, no extreme outliers remain in the dataset.



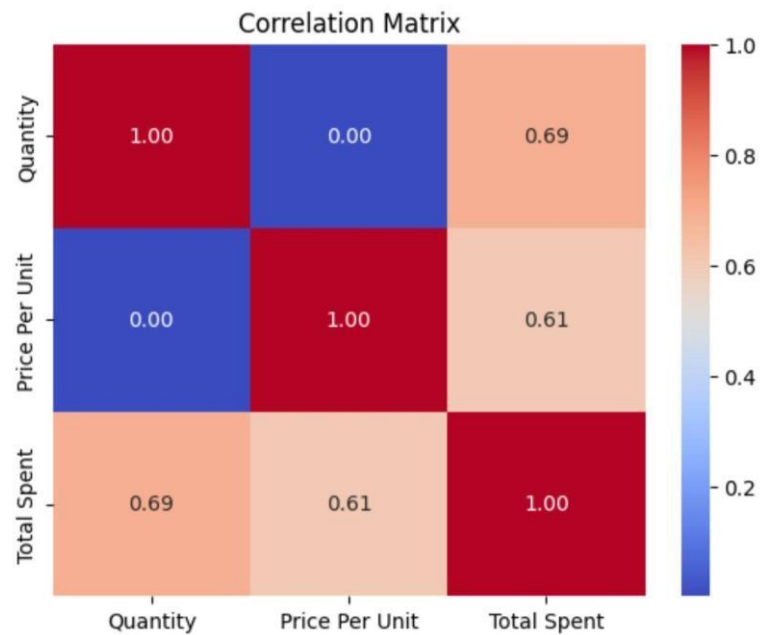
Visualization 3 – Quantity vs Total Spent (Scatter Plot)

- A clear positive relationship exists between quantity purchased and total amount spent.
- A correlation of 0.69 confirms a moderate-to-strong relationship.
- As expected, purchasing more items leads to a higher total bill.



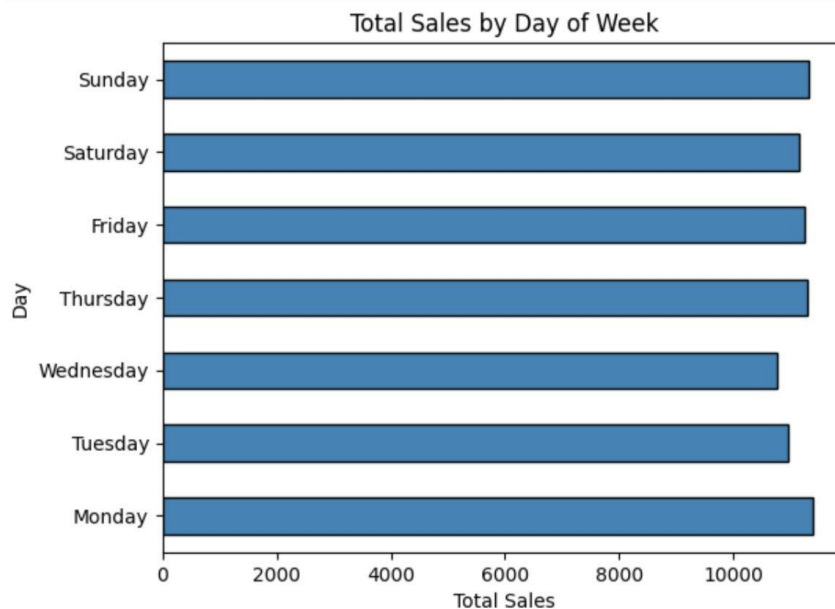
Visualization 4 – Correlation Matrix (Heatmap)

- Quantity and Total Spent: moderate positive correlation (0.69).
- Price Per Unit vs Quantity: zero correlation (0.00).
- Customers purchase the same quantity regardless of unit price — indicating price-insensitive behavior.



Visualization 5 – Total Sales by Day of Week (Bar Chart)

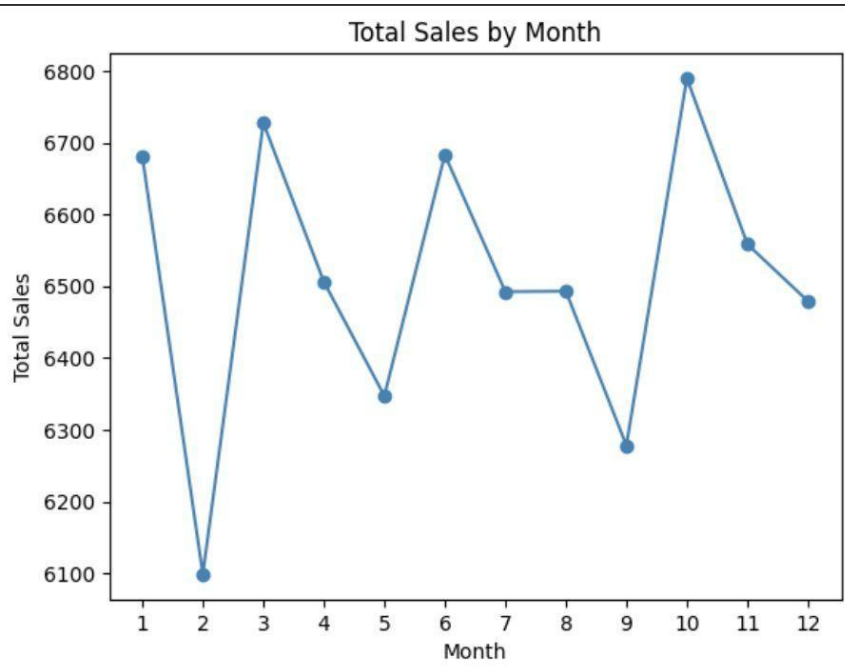
- Weekends (Friday, Saturday, Sunday) record the highest total sales.
- Customers are more likely to visit the cafe on weekends.
- Recommendation: increase staffing and stock levels on weekends to meet demand.



Visualization 6 – Total Sales by Month (Line Chart)

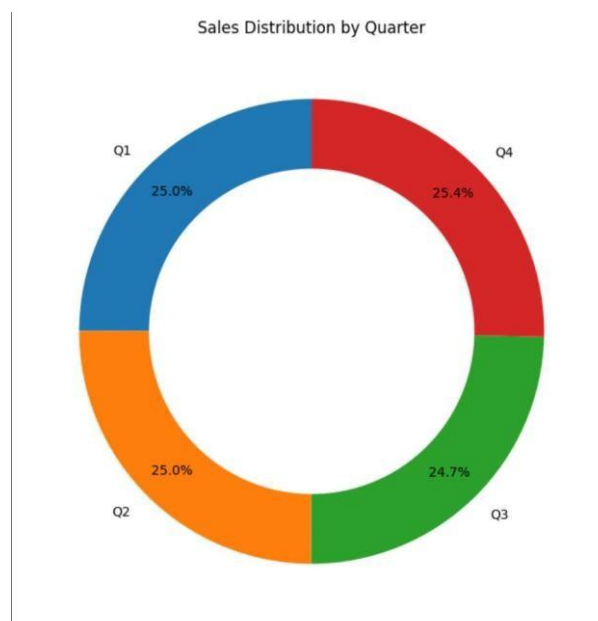
- October is the best-performing month (~6,800 in total sales).

- February is the weakest month (~6,100 in total sales).
- Sales fluctuate across the year with no consistent seasonal pattern.
- Recommendation: launch promotional campaigns in February to boost revenue.



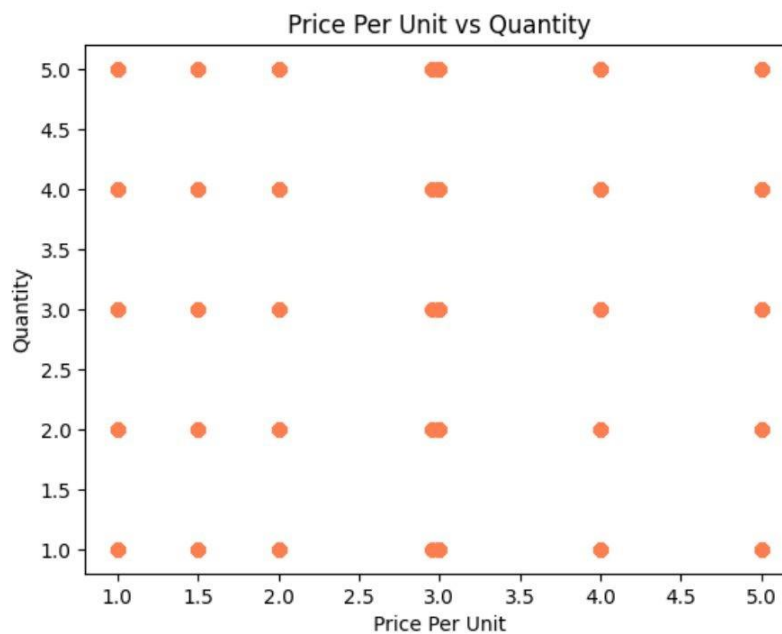
Visualization 7 – Sales Distribution by Quarter (Pie Chart)

- Sales are nearly evenly distributed across all four quarters.
- No single quarter dominates revenue, reflecting a stable year-round business.



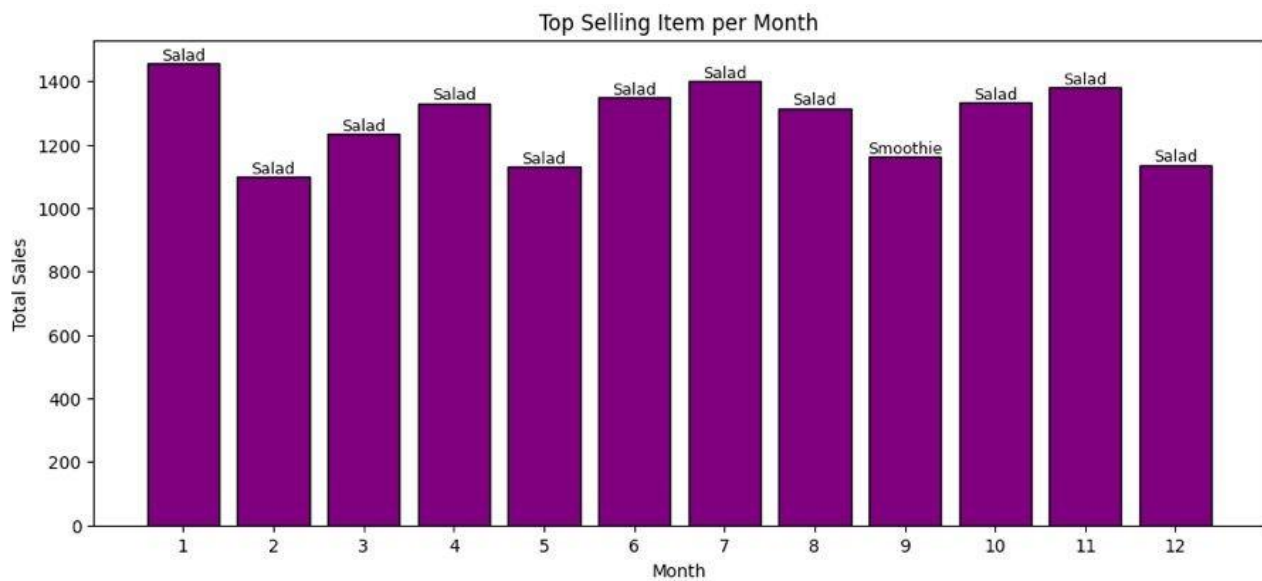
Visualization 8 – Price Per Unit vs Quantity (Scatter Plot)

- No visible correlation between price and quantity purchased (correlation = 0.00).
- Customers buy the same amount regardless of the price per unit.
- This reinforces the finding that customers in this cafe are not price sensitive.



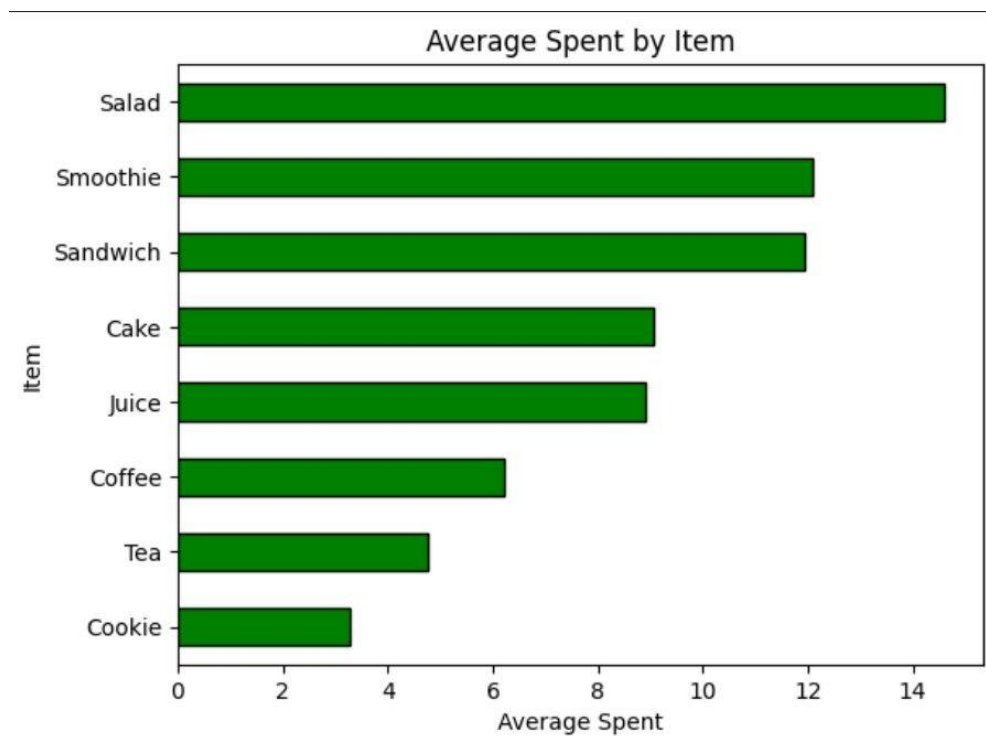
Visualization 9 – Top Selling Item per Month (Bar Chart)

- Salad is the top-selling item in almost every month of the year.
- Smoothie occasionally leads in certain months.
- Recommendation: ensure consistent stock of Salad ingredients throughout the year.



Visualization 10 – Average Spent by Item (Horizontal Bar Chart)

- Salad generates the highest average spending per transaction.
- Cake has the lowest average spending among all menu items.
- Recommendation: promote high-value items like Salad and Smoothie to increase average transaction value.



Chapter 3

Statistical Descriptions

3.1 Descriptive Statistics

The following statistical summary was generated using `df_clean.describe()` to explore the numerical columns of the cleaned dataset:

Statistic	Quantity	Price Per Unit	Total Spent
Count	8,788	8,788	8,788
Mean	3.02	2.95	8.89
Std Deviation	1.42	1.24	5.78
Min	1.00	1.00	1.00
25% (Q1)	2.00	2.00	4.00
50% (Median)	3.00	3.00	8.00
75% (Q3)	4.00	4.00	12.00
Max	5.00	5.00	24.00

3.2 Key Observation

- The average Total Spent is approximately 8.89, indicating moderate and consistent spending per visit.
- Quantity and Price Per Unit both range from 1 to 5, reflecting a limited but well-defined menu price range.
- The standard deviation for Total Spent (5.78) shows moderate variability in customer spending patterns.

- The median Total Spent (8.00) is close to the mean (8.89), suggesting an approximately symmetric distribution with no heavy skew.
- $Q1 = 4$ and $Q3 = 12$ for Total Spent, giving an IQR of 8 — used to detect and handle outliers during data cleaning.

Chapter 4

Data Quality & Cleaning

4.1 Data Quality Issues Identified

The following data quality issues were identified and classified across the dataset:

Issue Type	Column(s) Affected	Description
Incomplete	Item, Quantity, Transaction Date	Missing values in critical columns — rows dropped
Incomplete	Price Per Unit, Total Spent	Missing numerical values — filled with column mean
Incomplete	Payment Method	2,579 missing values — filled with mode
Incomplete	Location	3,265 missing values — filled with mode
Noisy	Total Spent	232 outliers detected beyond IQR bounds (upper = 24)
Inconsistent	Item	Mixed casing: 'Unknown', 'UNKNOWN' — standardized with str.title()
Intentional	Item, Payment Method, Location	Values labeled 'ERROR' replaced with 'Unknown'
Inconsistent	Transaction Date	Stored as string — converted to datetime64 format

4.2 Data Cleaning Performed

The following cleaning steps were applied sequentially to produce a clean, analysis-ready dataset:

Step 1 – Data Type Conversion

- Converted Quantity, Price Per Unit, and Total Spent to numeric using `pd.to_numeric()` with `errors='coerce'`.
- Converted Transaction Date to datetime format using `pd.to_datetime()` with `errors='coerce'`.

Step 2 – Remove Rows with Critical Missing Values

- Dropped all rows where Item, Quantity, or Transaction Date were null.
- These columns are essential for any meaningful analysis and cannot be imputed.
- Result: dataset reduced from 10,000 to 8,788 records.

Step 3 – Fill Missing Numerical Values

- Price Per Unit: filled missing values with the column mean to avoid introducing bias.
- Total Spent: filled missing values with the column mean.

Step 4 – Fill Missing Categorical Values

- Payment Method and Location: filled using the mode (most frequent value).
- This approach was chosen over 'Unknown' to avoid creating a large artificial category that would distort frequency-based analysis.

Step 5 – Handle Outliers in Total Spent

- Calculated IQR bounds: $Q1 = 4$, $Q3 = 12$, $IQR = 8$.
- Lower bound = $Q1 - 1.5 \times IQR = -8$ | Upper bound = $Q3 + 1.5 \times IQR = 24$.
- Detected 232 outlier values above 24.
- Applied `df.clip(lower, upper)` to cap values within acceptable range without losing records.

Step 6 – Standardize Inconsistent Text Values

- Applied `str.strip().str.title()` to the Item column to unify all casing variations into a consistent format (e.g., 'UNKNOWN' and 'Unknown' → 'Unknown').
- Replaced all remaining 'ERROR' values in Item, Payment Method, and Location with 'Unknown'.

Step 7 – Save Cleaned Dataset

- Exported the fully cleaned dataset to `clean_cafe_sales.csv` using `df_clean.to_csv()` with `index=False`.
- Final dataset: 8,788 records, 8 columns, zero null values.